Data visualization with ggplot2

POLI 210 - Seminar 9

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Questions for this seminar

- How persistent is a district's Democratic vote?
- How do House and presidential voting line up?
- What does the distribution of congressional ideology look like within parties?
- How can we display the summaries and fitted results in a published paper?

Reading: PSD Chapter 7. Today we build a few useful plots, one layer at a time.

What does one row mean?

Data	One row	Important limitation
`elections`	Retained district-election	Selected extract, not all races
`matched`	Retained district-election with a unique member match	Following Congress; not voter ideology
`house`	Member record, 114th House	May include replacements

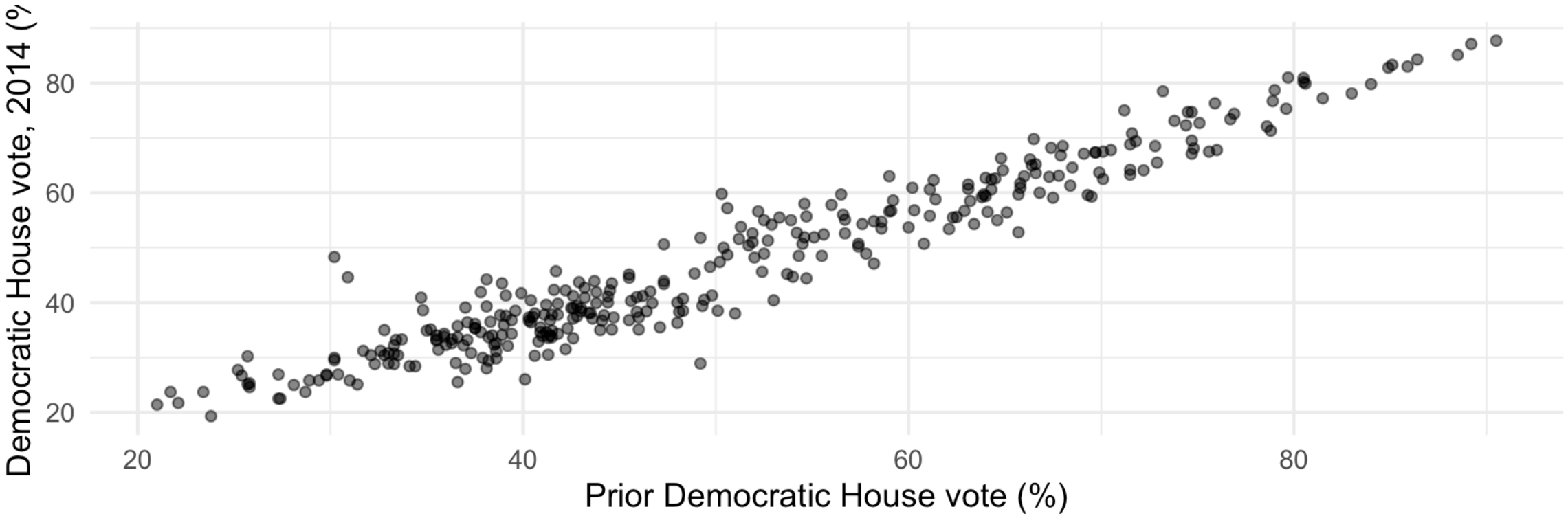
The election file has 9,337 rows, 1946–2014. Its 2014 subset has 328 rows. Do not turn these counts into national seat totals.

A small grammar of graphics

```
ggplot(data = e2014, mapping = aes(x = dvp, y = dv)) +
  geom_point() +
  labs(x = "Prior Democratic House vote (%)",
       y = "Democratic House vote, 2014 (%)")
```

- **Data:** which rows and columns?
- **Mapping:** which variables control positions, colors, or other features?
- **Geometry:** points, bars, lines, boxes …
- **Labels/scales/theme:** explain and refine the display.

Each point is a district-election



`dvp`: prior House election. `dv`: current House election. Both are percentages.

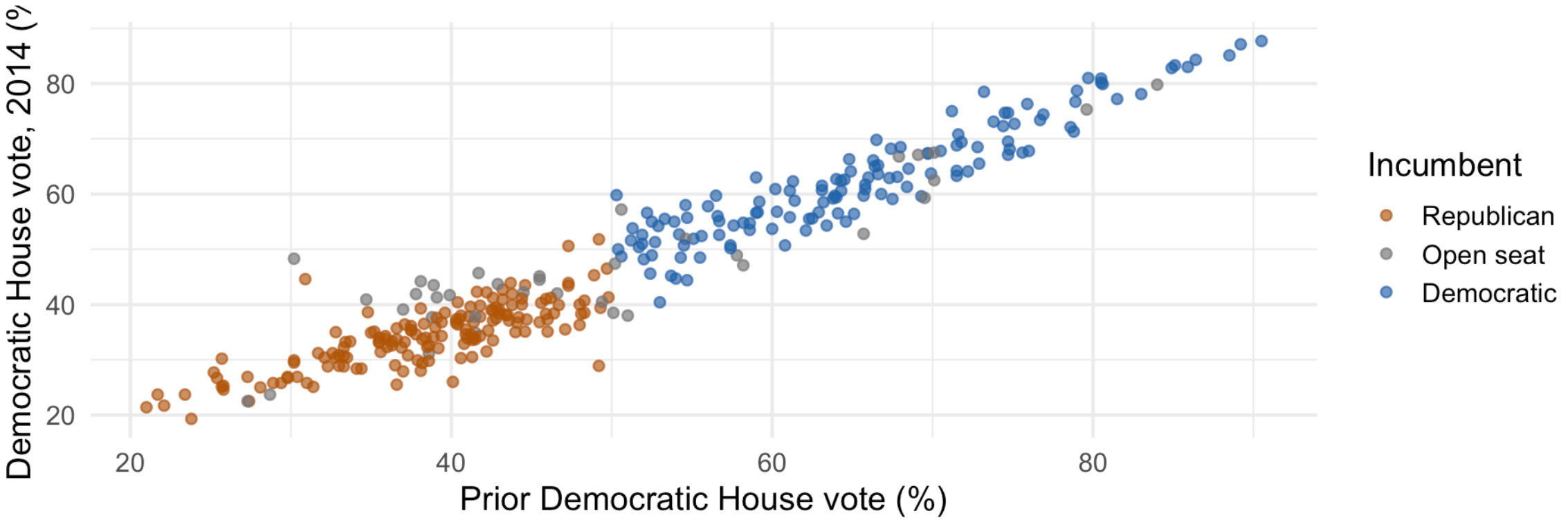
Add a variable through a mapping

```
ggplot(e2014, aes(dvp, dv)) +
  geom_point(aes(color = incumbent), alpha = 0.7) +
  scale_color_manual(values = inc_colors)
```

`aes(color = incumbent)` says: look up each row's incumbent category and map it to color.

`alpha = 0.7` says: make **every** point partly transparent.

Incumbency provides additional context



This is a descriptive pattern, not an estimate of the causal effect of incumbency.

Mapping versus setting

```
geom_point(aes(color = incumbent)) # Variable -> colors + legend
geom_point(color = "blue")         # One fixed color
geom_point(aes(color = "blue"))    # A category called "blue"!
```

Inside `aes()`: map a variable (or expression) to a visual feature. Outside `aes()`: set a constant appearance.

`color` controls points/lines/outlines; `fill` controls interiors of bars and ribbons.

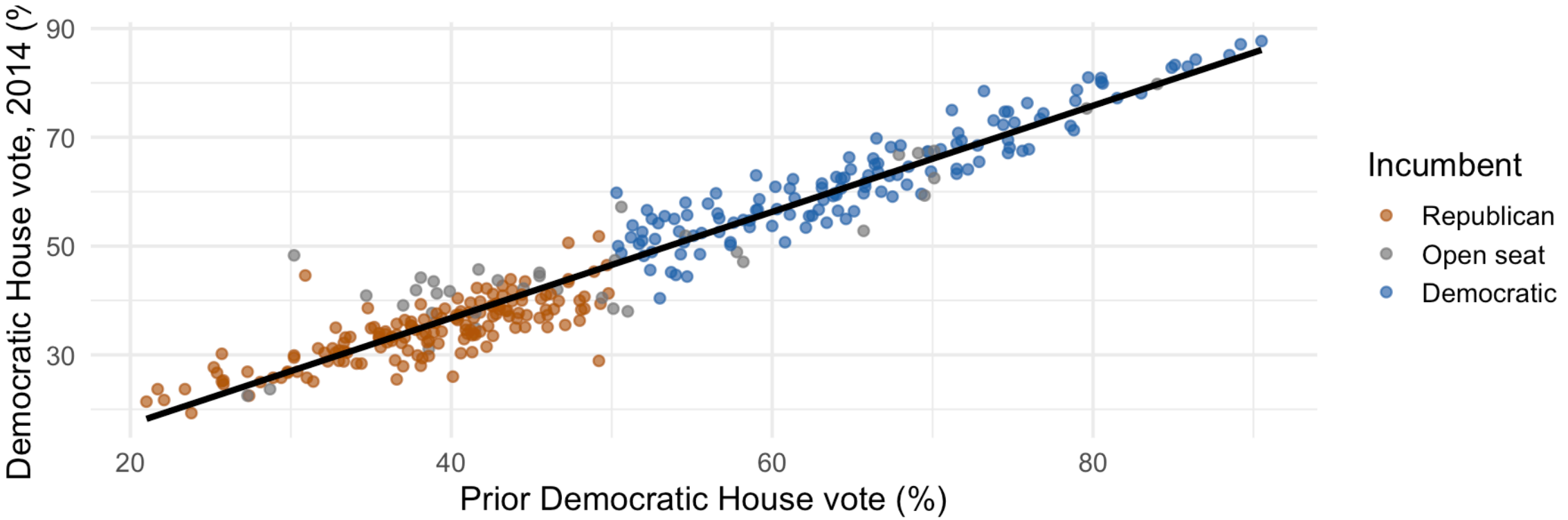
Layers inherit mappings

```
ggplot(e2014, aes(dvp, dv)) +
  geom_point(aes(color = incumbent), alpha = 0.7) +
  geom_smooth(method = "lm", formula = y ~ x,
              se = FALSE, color = "black")
```

Color is mapped **only in the point layer**. The fitted line uses all rows. Putting a categorical color mapping in `ggplot()` would normally fit a line for each group.

We specify a straight-line fit explicitly. Regression interpretation comes later.

One fitted line, several point colors



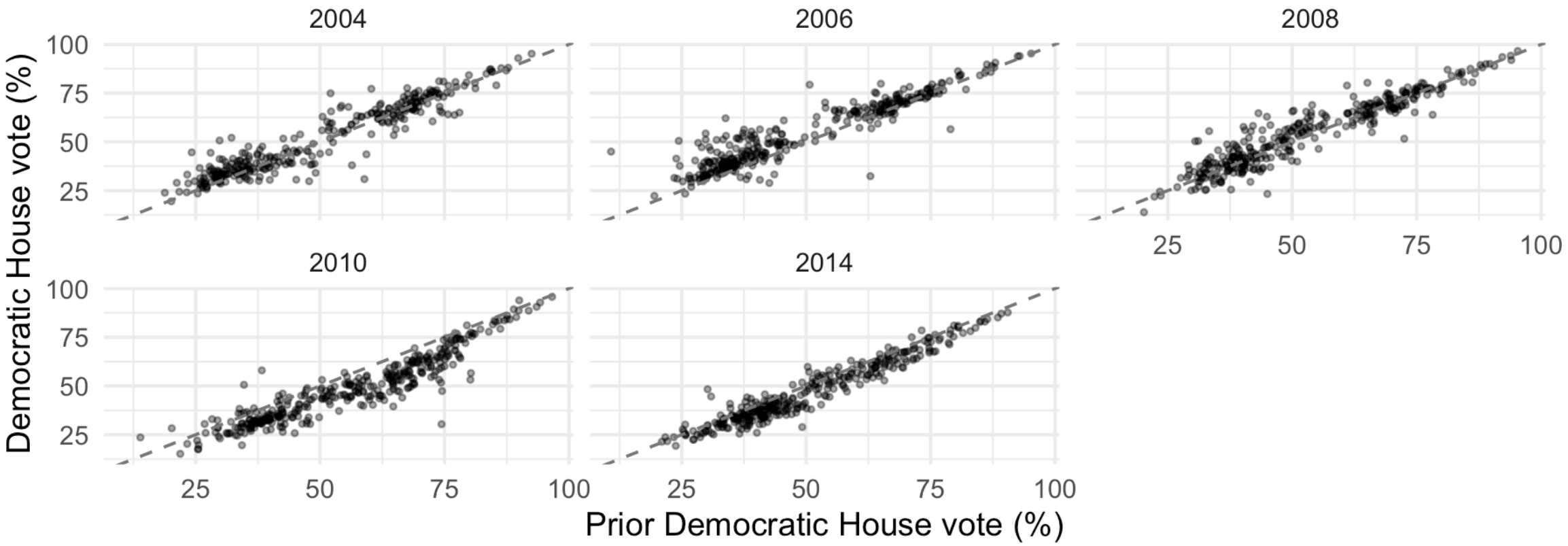
The line summarizes the relationship; it does not establish causation.

Facets repeat a plot for subsets

```
ggplot(recent, aes(dvp, dv)) +
  geom_point(alpha = 0.4) +
  geom_abline(slope = 1, intercept = 0, linetype = 2) +
  facet_wrap(~year, nrow = 2)
```

The dashed line marks unchanged Democratic vote share, not a fitted relationship. Fixed axes make panels comparable. These are the five latest years in our extract.

Comparing elections on the same axes



In which panels do observations tend to fall below the equality line?

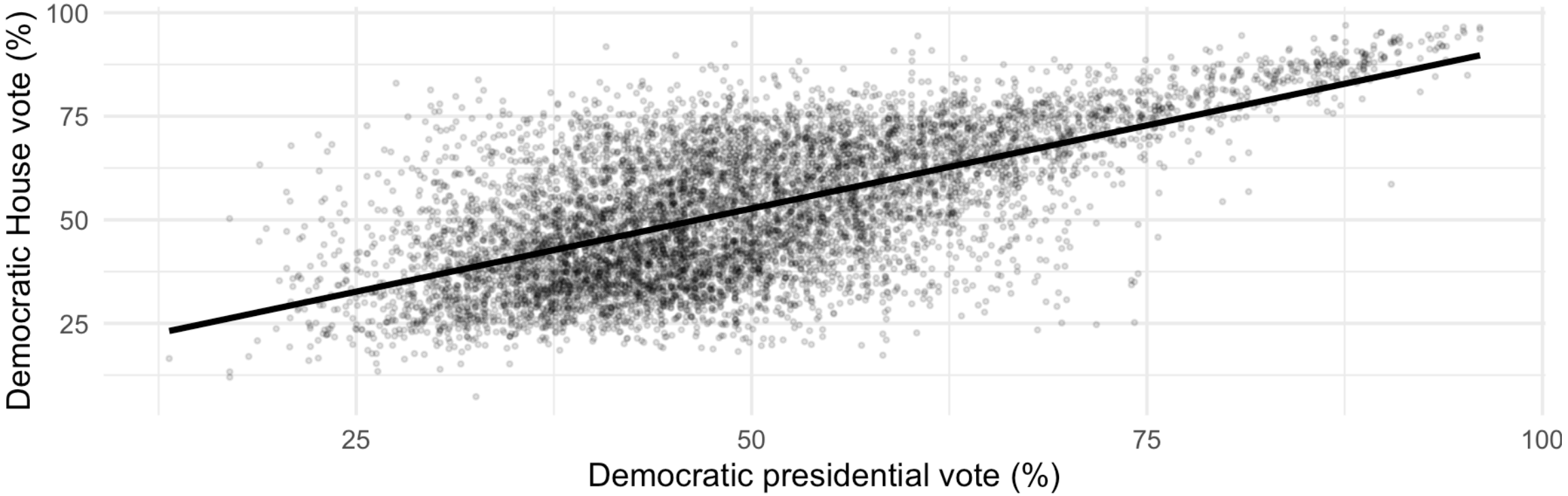
House vote and presidential vote

`dpres` measures district Democratic two-party presidential vote (%). For a midterm House election, it refers to the preceding presidential election.

```
pres <- elections |> filter(!is.na(dpres))
ggplot(pres, aes(dpres, dv)) +
  geom_point(alpha = 0.14, size = 0.7) +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE)
```

There are 8,212 usable rows. Transparency helps with overlapping points.

Pooled elections: a strong descriptive pattern



Pooling years can conceal change. A fitted line is not a substitute for checking subsets.

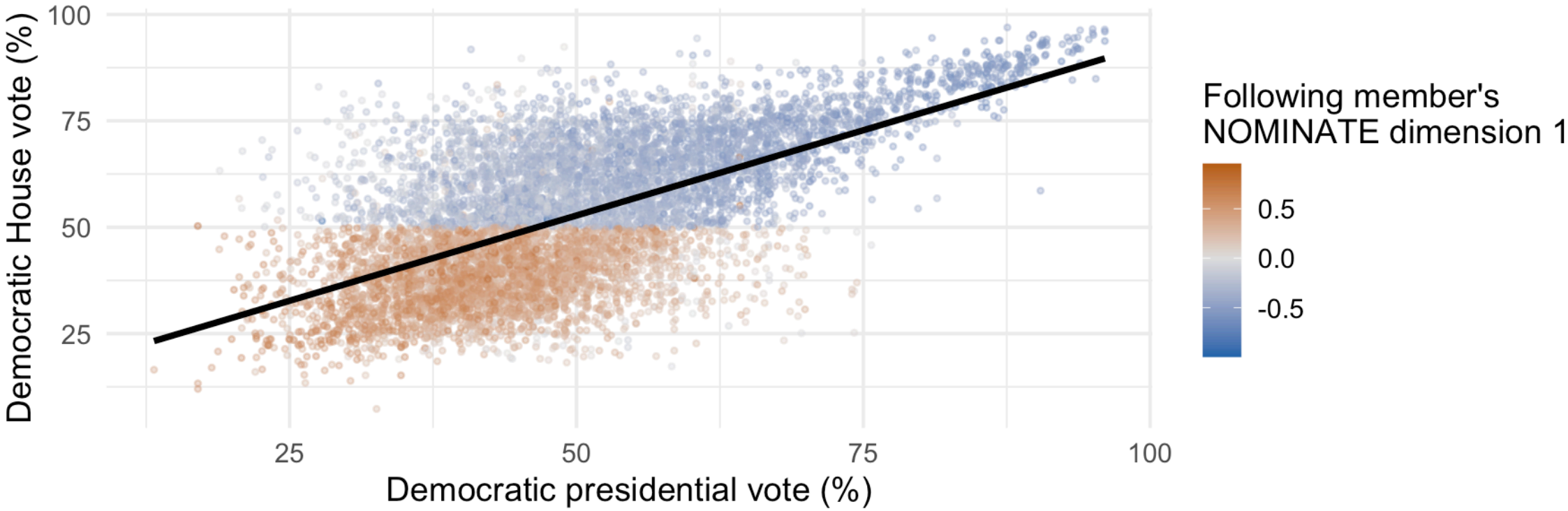
Matching elections to member ideology

The prepared file matches an election to the **following Congress**: 2014 → 114th Congress (2015–2017), using state and district.

- The color plot includes only keys with exactly one House member record.
- Ambiguous or unmatched keys are retained in the data, not guessed.
- Of 8,212 rows with `dpres`, 8,061 have a usable unique match.

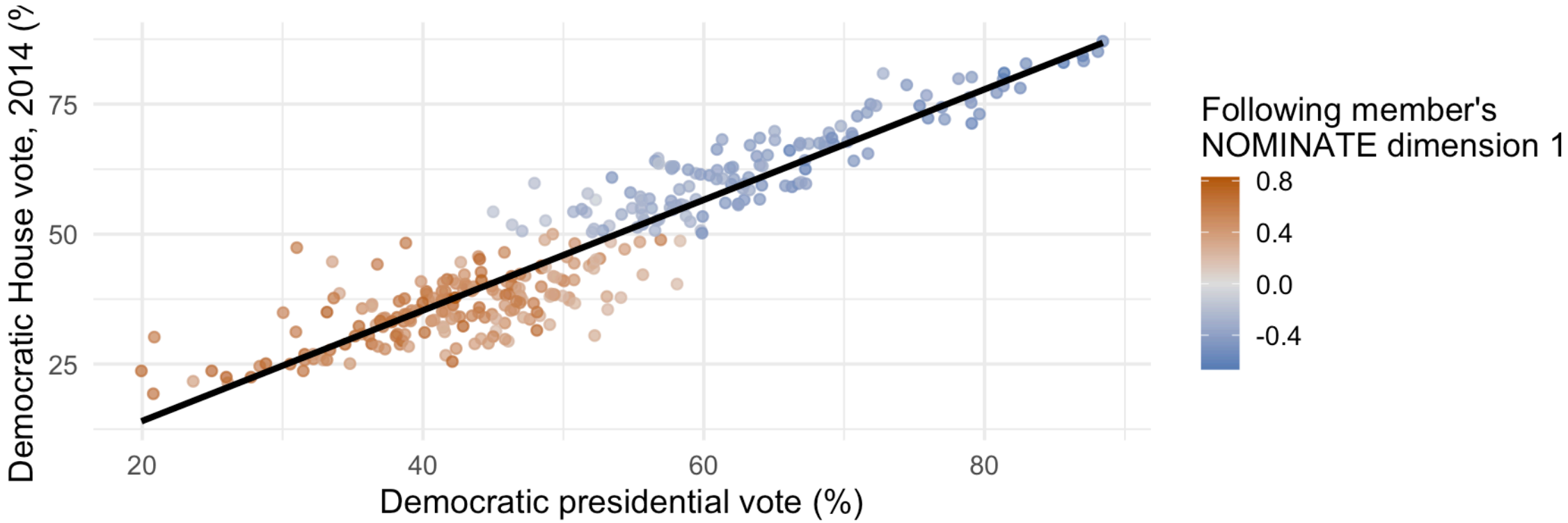
NOMINATE dimension 1 describes the member's roll-call ideology. It is **not** a district opinion measure or a predictor known before that election.

Continuous color: member ideology



Blue = more liberal; orange = more conservative. The black fit ignores the color mapping.

Presidential vote and member ideology in 2014



2014 only: easier to see the observations; still a selected, descriptive comparison.

Interpreting the House-vote scatterplot

With prior House vote (`dvp`) on x and current House vote (`dv`) on y:

- **Equality line (`dv = dvp`):** points above it indicate an increase in Democratic vote share.
- **Incumbent colors:** distinguish districts with Republican incumbents, Democratic incumbents, and open seats.
- **Facets by election:** reveal whether patterns persist or change across years.

Why might a line fitted to all districts differ from separate lines fitted within incumbent categories?

Break

Next: move from individual observations to distributions and summaries.

The key question remains: **What quantity does the height or position represent?**

Count rows or plot an existing value?

Goal	Geometry	Input
Count cases in categories	`geom_bar()`	Individual rows
Count cases in numeric intervals	`geom_histogram()`	Individual rows
Show a calculated mean/ proportion	`geom_col()`	Summary rows

`geom_bar()` counts by default. `geom_col()` uses your y values. Using the wrong one changes the question the plot answers.

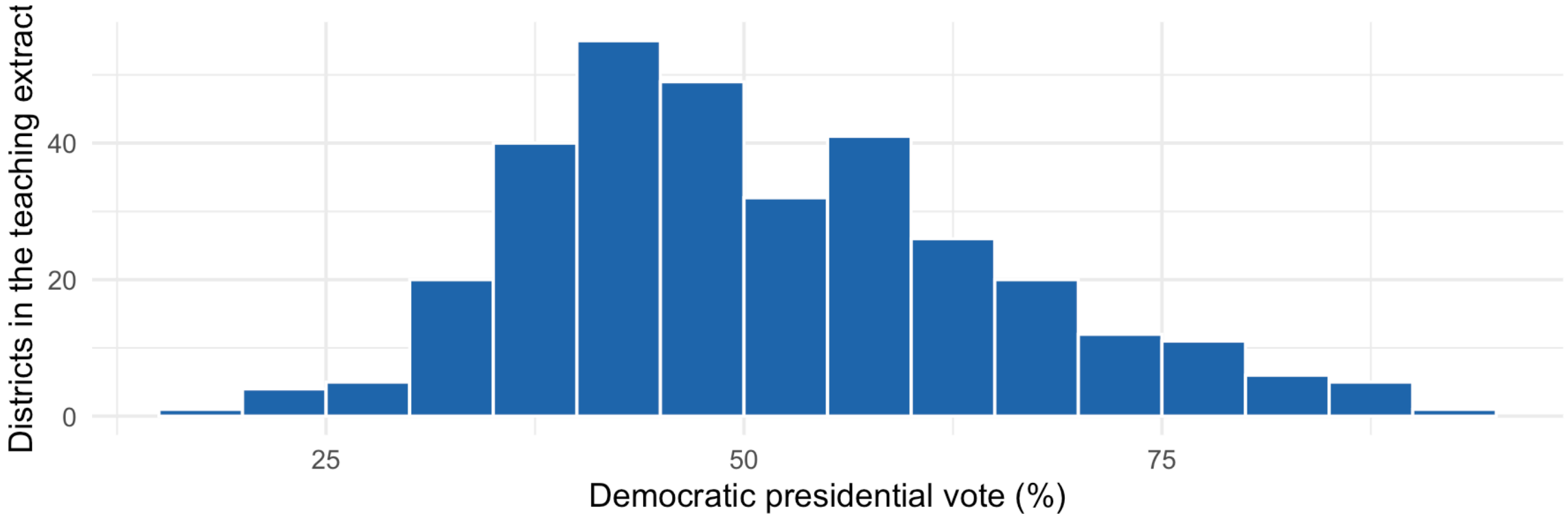
Histograms: numeric intervals

```
ggplot(e2014, aes(dpres)) +
  geom_histogram(binwidth = 5, boundary = 0,
                 fill = "steelblue", color = "white")
```

Bin width is **5 percentage points**. Each height counts retained districts in that interval.

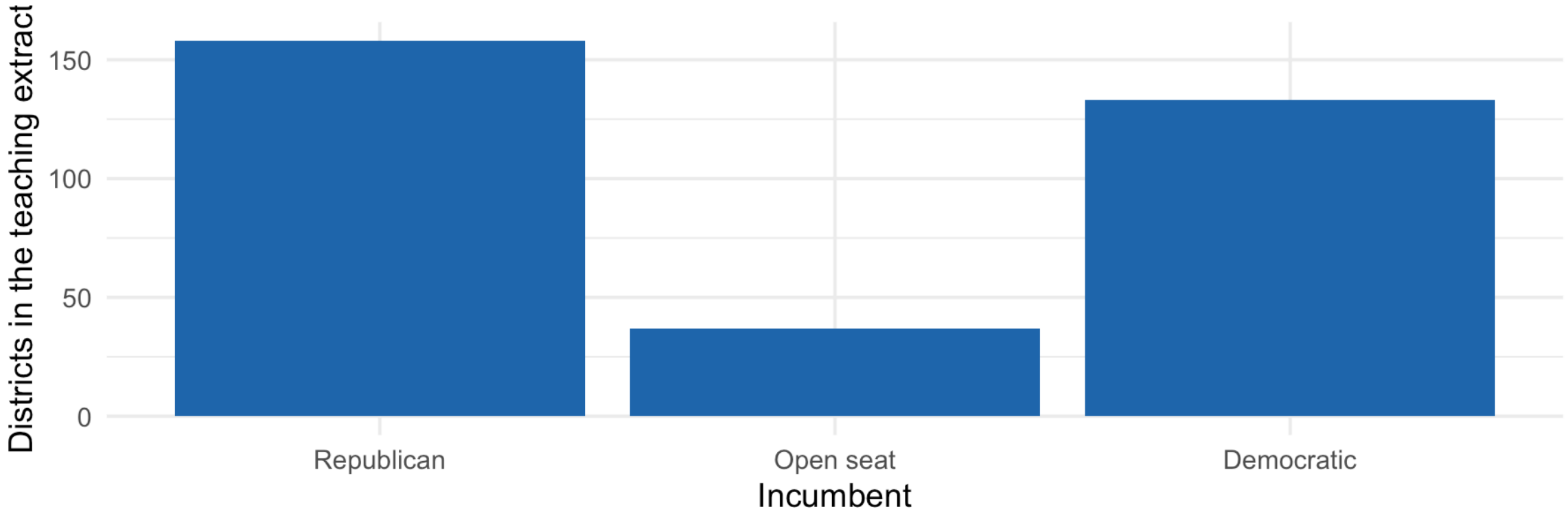
Bin widths of 2, 5, and 10 show different levels of detail using the same observations. For a density-scaled histogram, map `y = after_stat(density)` instead.

Distribution of presidential vote, 2014 districts



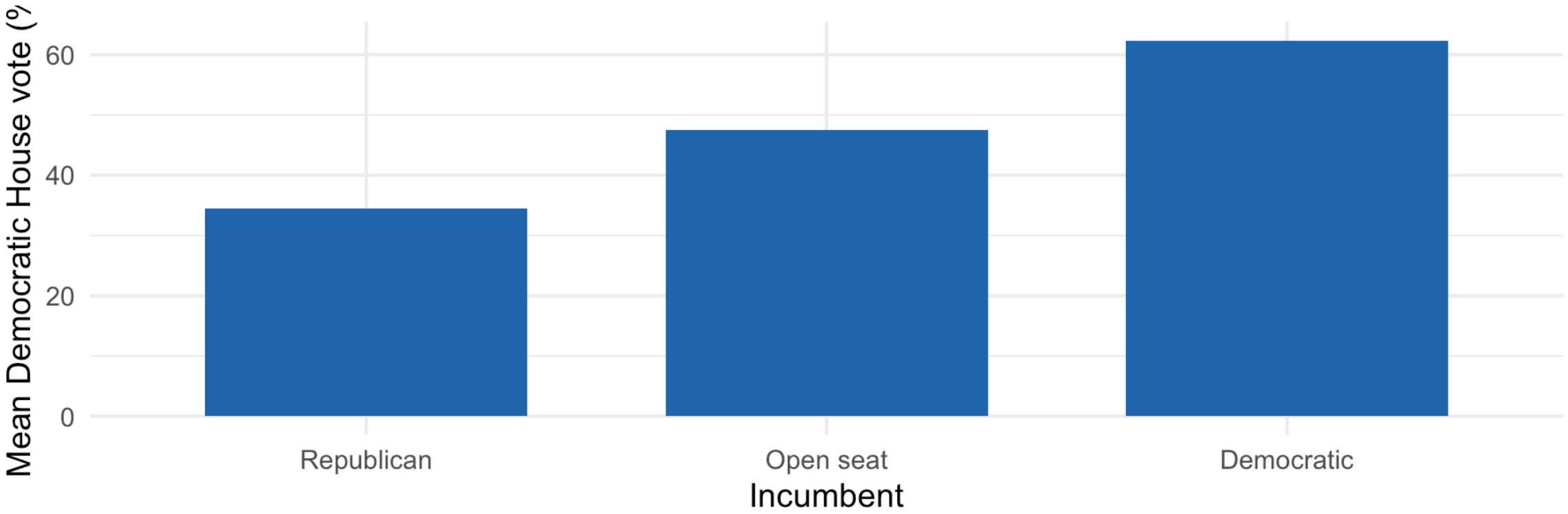
These are district counts, not counts of voters.

Bars for counts of categories



`ggplot(e2014, aes(incumbent)) + geom_bar()` counts the rows in each category.

Bars for calculated means



`ggplot(means, aes(incumbent, mean_dv)) + geom_col()` uses the summary table's y column.

Reshape first, then map

Our original data have separate columns `dv` and `dvp`. For grouped bars, one column should name the election; another should contain the vote.

```
vote_long <- e2014 |>
  select(incumbent, dv, dvp) |>
  pivot_longer(c(dv, dvp),
               names_to = "election", values_to = "vote")
```

One row now represents a district's vote at **one of two times**. This does not create additional independent districts.

Calculate before drawing

```
vote_summary <- vote_long |>
  group_by(incumbent, election) |>
  summarise(mean_vote = mean(vote), n = n(),
            se = sd(vote) / sqrt(n()), .groups = "drop")
```

One row now represents an incumbent category × election summary.

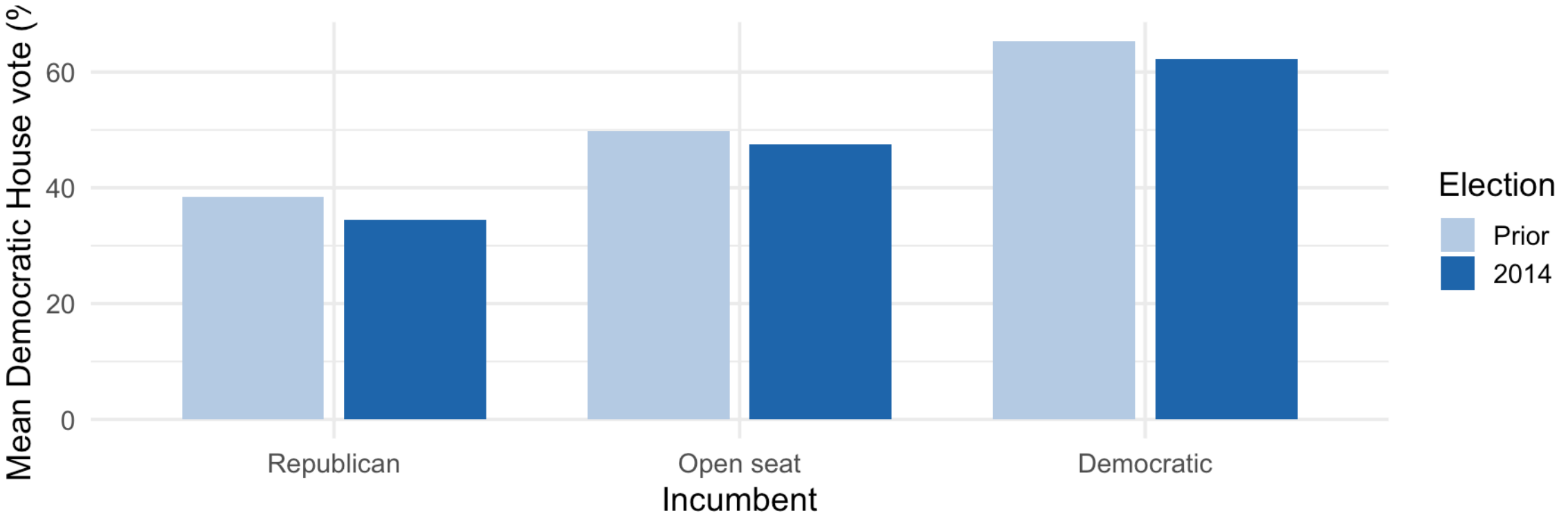
The `se` formula is an **illustrative IID calculation**. This selected election extract does not justify treating it as a representative random sample.

The summary table behind the plot

incumbent	election	mean_vote	n	se
Republican	Prior	38.51	158	0.52
Republican	2014	34.40	158	0.46
Open seat	Prior	49.90	37	2.40
Open seat	2014	47.56	37	2.16
Democratic	Prior	65.35	133	0.86
Democratic	2014	62.31	133	0.89

`sd(vote)` describes variation across districts. `sd(vote)/sqrt(n)` estimates the standard error of a mean under IID sampling. They answer different questions.

Side by side, not stacked



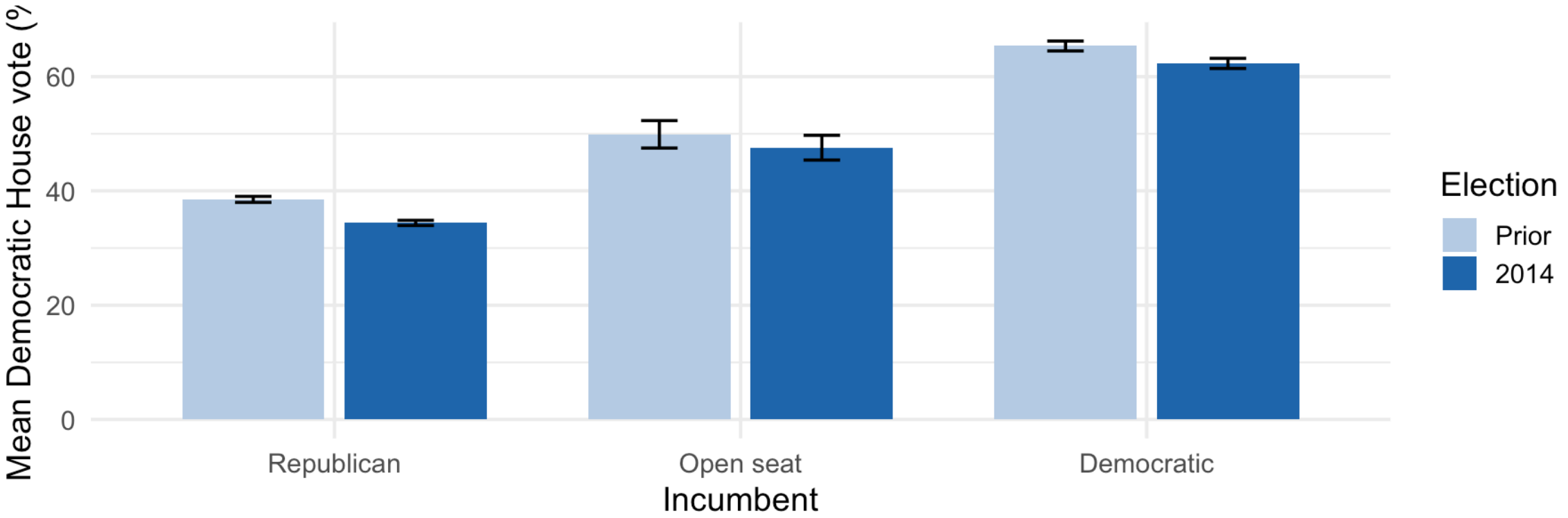
Map election to `fill`; use `position_dodge(width = 0.8)`. Keep bar baselines at zero.

Add uncertainty to the same positions

```
dodge <- position_dodge(width = 0.8)
ggplot(vote_summary, aes(incumbent, mean_vote, fill = election)) +
  geom_col(position = dodge) +
  geom_errorbar(aes(ymin = mean_vote - se,
                    ymax = mean_vote + se),
                position = dodge, width = 0.18)
```

Use the same dodge for bars and whiskers. Whisker width is a graphical choice; interval height comes from the calculated endpoints.

Interpreting the whiskers



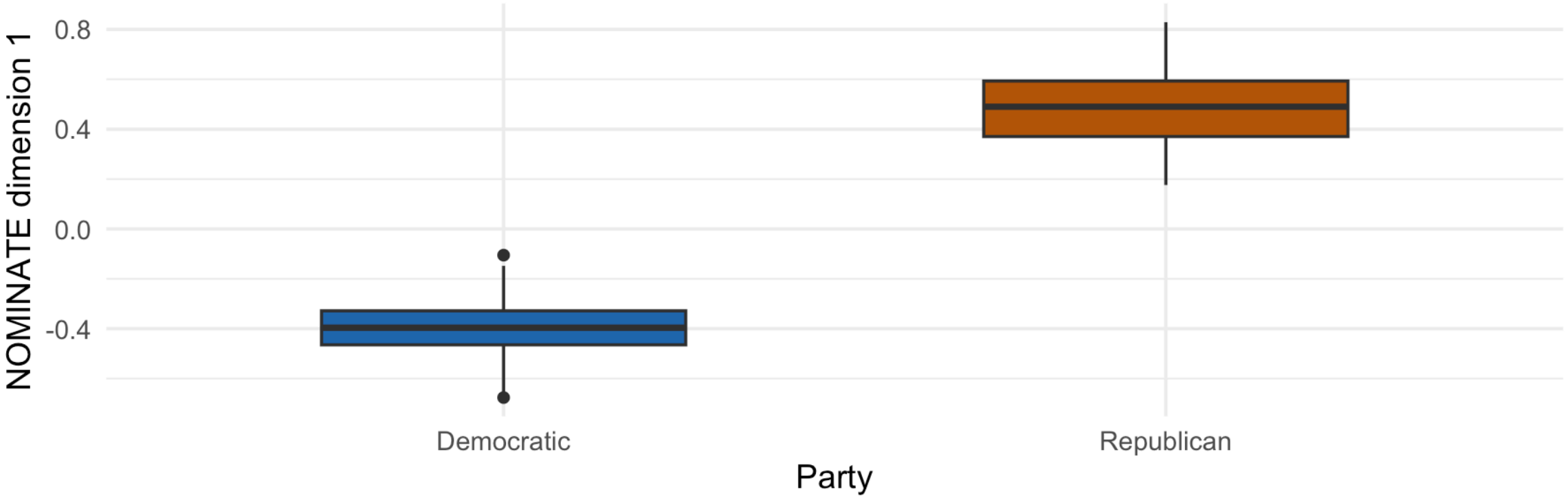
Here: mean ±1 illustrative IID SE. Not ±1 SD; not a 95% confidence interval. B4 supplies bootstrap endpoints for Figure 1 instead.

A boxplot summarizes a distribution

- Center line: median.
- Box: 25th to 75th percentile (the interquartile range, IQR).
- Whiskers: most extreme observations within 1.5 IQR of the box edges.
- Points beyond the whiskers: observations, not automatically errors.

A box is **not a confidence interval for the mean**. Compare spread, asymmetry, and location, not just one number.

Member ideology within parties



`geom_boxplot()` calculates the summaries from member-level rows. 114th House, Democratic and Republican member records with available scores.

Lines and ribbons can use supplied results

Suppose `predictions` has `x`, `estimate`, `lower`, `upper`, and `series`.

```
ggplot(predictions, aes(x, estimate)) +
  geom_ribbon(aes(ymin = lower, ymax = upper,
                  fill = series, group = series), alpha = 0.2) +
  geom_line(aes(color = series))
```

Draw the ribbon first so it does not cover the line. These rows are **predictions at a grid of x values**, not new observations.

Different layers can use different tables

```
ggplot() +
  geom_point(data = observed, aes(x, outcome), alpha = 0.5) +
  geom_line(data = predictions, aes(x, estimate))
```

Start with an empty `ggplot()` and supply each layer's `data` and mappings. Otherwise a layer may inherit a column name that its own table does not have.

This is the pattern for B4′s observed experience points and supplied fitted curves.

Finish and save a reproducible plot

```
p <- ggplot(e2014, aes(dvp, dv)) +
  geom_point(alpha = 0.5) +
  labs(x = "Prior Democratic vote (%)", y = "2014 Democratic vote (%)") +
  theme_minimal(base_size = 13)
p
ggsave("house-vote.png", plot = p, width = 7, height = 4.5, dpi = 300)
```

In Quarto, printing `p` includes it in the report. `ggsave()` is for a separate file. Check labels and units at the size the reader will actually see.

B4: seven panels, three kinds of input

Figure	What you prepare or receive	Plot
1A–C	Group means/proportions + bootstrap endpoints	Bars + whiskers
2A	Supplied patience predictions	Lines + 95% ribbons
2B	Supplied differences relative to K=0	Bars + SE whiskers
3A–B	Elite observations + supplied log-experience fits	Points + lines + ribbons

Download the archive files; do not join overlapping samples by row number. Each source table determines the quantity shown by a panel's heights or positions.

Optional: a Normal Q–Q plot

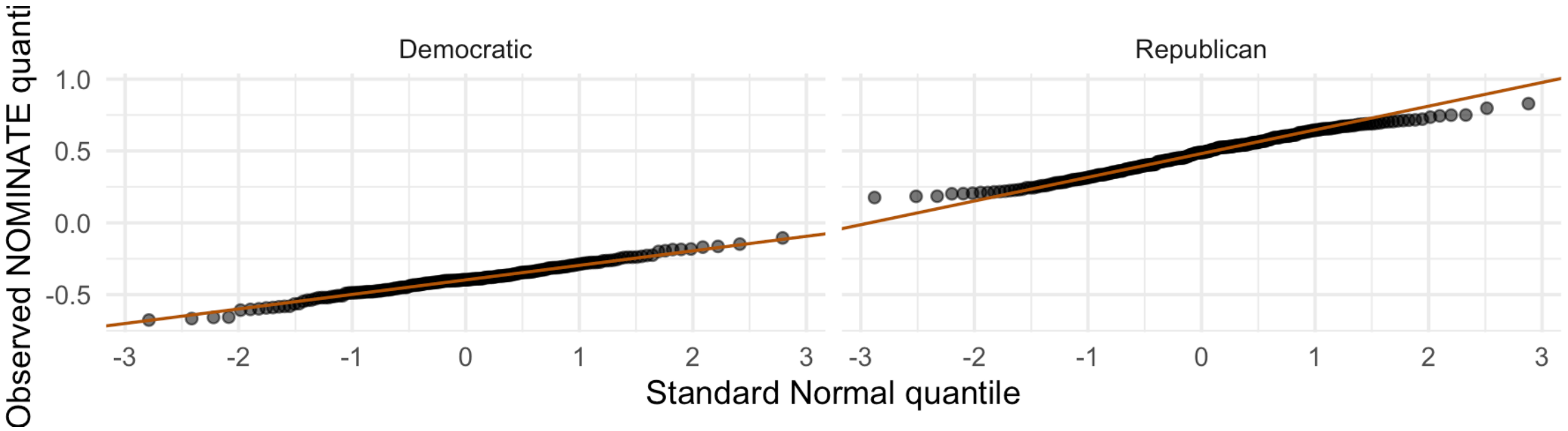
Compare the observed quantile at each probability with a Normal quantile at that probability.

If a distribution is approximately Normal, the points should follow roughly a straight line. The intercept and slope can reflect a nonzero mean and a nonunit SD.

The reference line passes through the first and third quartiles by default. Systematic curvature or tail departures are useful diagnostics, not a pass/fail test.

Optional: compare within parties

```
ggplot(house, aes(sample = nominate_dim1)) +
  stat_qq() + stat_qq_line() + facet_wrap(~party)
```



Optional: what a Q–Q plot does not establish

- NOMINATE scores need not be Normal for these descriptive plots to be useful.
- A mixture of parties can be non-Normal even when within-party shapes look simpler.
- A variable’s distribution is not the sampling distribution of its mean.

Reading and sources

- PSD Chapter 7: Data visualization.
- Jacobson (2015), election replication archive; selected teaching extract.
- Voteview member data documentation; frozen October 20, 2026 download.
- LeVeck et al. (2014), The role of self-interest in elite bargaining, *PNAS* 111(52): 18536–18541; replication data.
- ggplot2 reference: geoms, mappings, scales, and facets.